

Precalc Lesson 1.4 Homework

Name _____

1. Identify the domain and range of $y = \sqrt{x}$. Give your answer in interval notation.

2. Identify the parent function of f .
 - a. $f(x) = 2 - 11x + 23x^2$

 - b. $f(x) = 3|x - 4| + 8$

3. Use interval notation to describe the intervals on which $y = x^2$ is increasing, decreasing, and constant.

4. Let f be the identity function and g be the reciprocal function. Find $4f(-2) + g(-2)$.

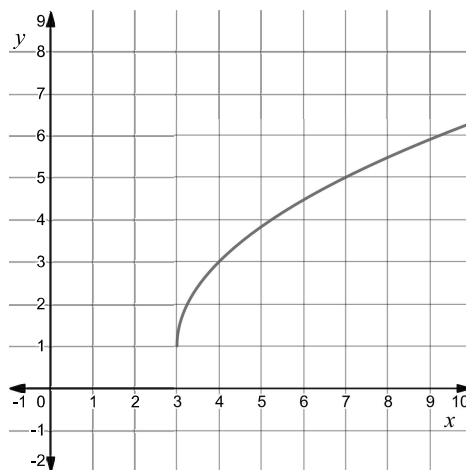
5. Which of the following parent functions is increasing over its entire domain?
 - A) $y = x^3$
 - B) $y = |x|$
 - C) $y = \frac{1}{x}$
 - D) $y = x^2$

6. Consider the parent functions $y = \frac{1}{x}$, $y = x^2$, $y = x^3$, and $y = x$.

a. Which of these functions has the greatest output at $x = \frac{1}{3}$?

b. Which of the functions has the least output at $x = \frac{1}{3}$?

7. The graph of f is shown. Identify the parent function of f .



8. Consider the parent functions $f(x) = x^2$ and $g(x) = x^3$. For some values of k , it is true that $f(k) > g(k)$. Find two possible values of k .

9. Decide if the statement is always, sometimes, or never true. Give a reason for your answer.

$$|x| \geq x \text{ for any real number, } x.$$

10. Let $g(x) = ax + b$, for some non-zero constants a and b . Describe the differences between the graph of g and its parent function. What is the domain and range of g ?